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TOPIC MODELLING

LDA, Gibbs sampling, Top2Vec and BERTopic

Sine Zambach
Assistant Professor, DIGI

Some of the slides are adapted from:

- David M. Blei, **Probabilistic Topic Models,** http://yosinski.com/mlss12/media/slides/MLSS-2012-Blei-Probabilistic-Topic-Models.pdf

- Nam Khanh Tran, Topic Models, 2014

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Hi – who am I?

- Sine Zambach, Assistant Professor at CBS, sz.digi@cbs.dk
  - Cand.scient. In Bioinformatics
  - PhD in logic and language in Computer Science
- Experience and research
  - Learning Analytics
  - NLP in evaluation comments, etc
  - Data visualization and management communication
  - Bioinformatics (Genetics, …)
  - Teaching introduction to R and Natural Language Processing

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Important info regarding report and your studies

Need more calculation power?

You can access our GPUs from Kiran Kumar Kocherla
kkk.digi@cbs.dk

You can also get access to high performance computing (HPC) through library/Kristoffer Gulmark Poulsen
kgp.lib@cbs.dk

Slides adapted from Jure Leskovec

Agenda

- Topic Modeling
- Latent Dirichlet Allocation
  w/ Gibbs sampling
- BERTopic

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What should you know after todays lecture?

- Understand – overall – what is topic modelling and how can it be used?
- Understand intuition behind LDA
  - How it generates topics
- Understand why and how we use gibbs sampling
  - To help with parameter estimation and how it does so
- Understand intuition behind BERTopic
  - How it generates topics and what are the pro's and con's compared to LDA? And ChatGPT

Slides adapted from Jure Leskovec

SO – WHAT IS TOPICS?

When will you need to find
”hidden” topics in text?

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Why would you want topics?

You want to learn something about a corpus - too big to read.

Trending topics on Reddit? - billions post daily.

Input all these documents to the computer and see what interesting patterns it finds?

Summary: What is Topic Modeling?

Unsupervised machine learning techniques (like clustering)

Type of statistical modelling for discovering the abstract “topics” that occur in a collection of documents.

Provides methods for automatically organizing, understanding, searching, and summarizing large electronic archives.

Useful for document clustering, organizing large blocks of textual data, information retrieval from unstructured text and feature selection.

Slides adapted from Jure Leskovec

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LATENT DIRICHLET ALLOCATION (LDA)

Blei et al, 2003

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Latent Dirichlet Allocation (LDA) - meaning

Latent	• Means hidden, something that is yet to be found.
Dirichlet	• Model assumes that the topics in the documents and the words in those topics follow a Dirichlet distribution.
Allocation	• Means giving something, which in this case are topics.

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http://www.cs.columbia.edu/~blei/papers/Blei2012.pdf

Intuition: LDA is a generative model

text is "generated" by the topics

Each **topic** is a distribution over words

Each **document** is a mixture of corpus-wide topics

Each **word** is drawn from one of those topics

Topics

```
gene      0.04
dna       0.02
genetic   0.01
...
```

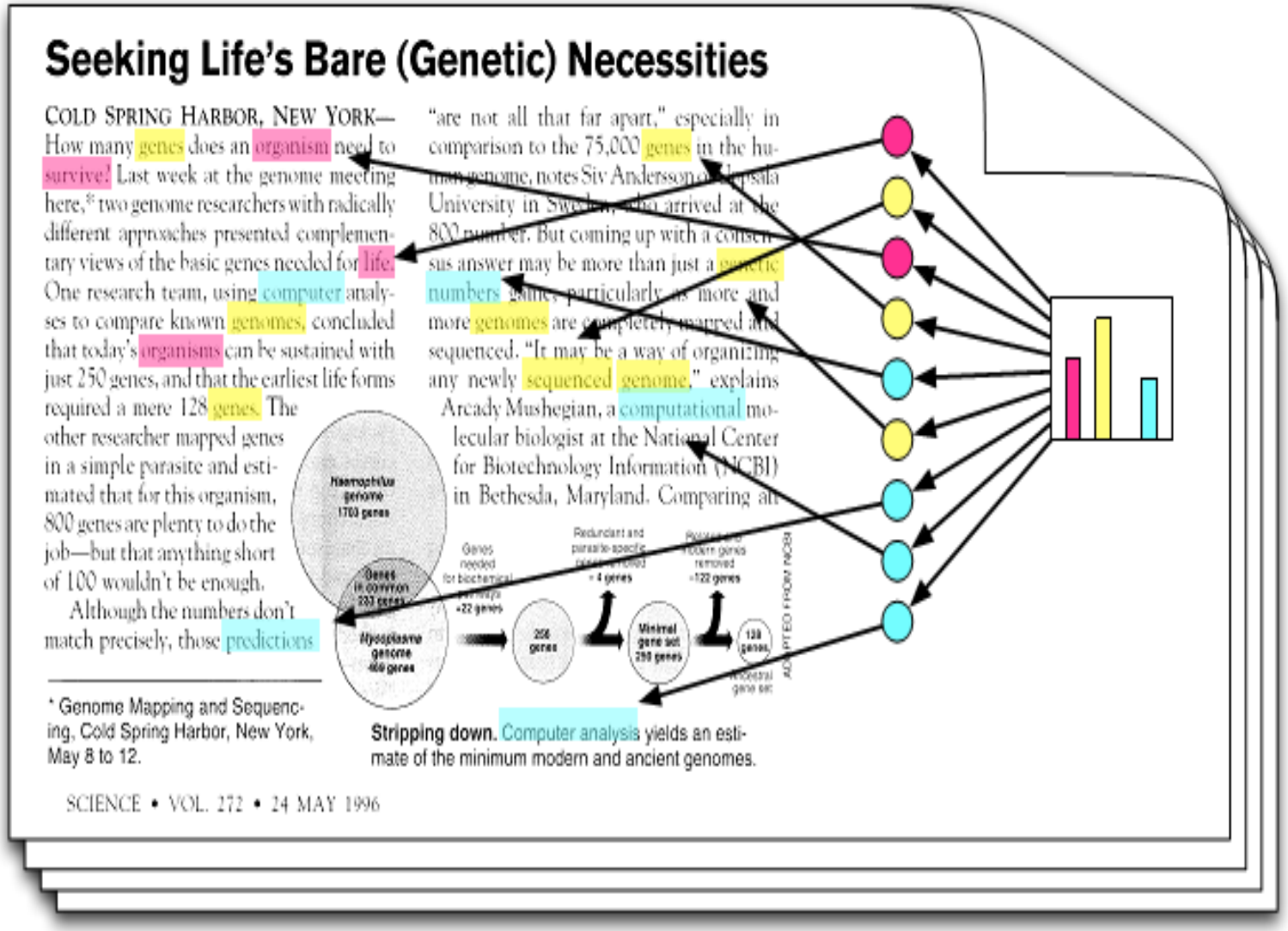
```
life      0.02
evolve    0.01
organism  0.01
...
```

```
brain     0.04
neuron    0.02
nerve     0.01
...
```

```
data      0.02
number    0.02
computer  0.01
...
```

Documents

Topic proportions and assignments



Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions "are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

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SCIENCE • VOL. 272 • 24 MAY 1996

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Latent Dirichlet Allocation (LDA) in short

Treats documents as bags of words.

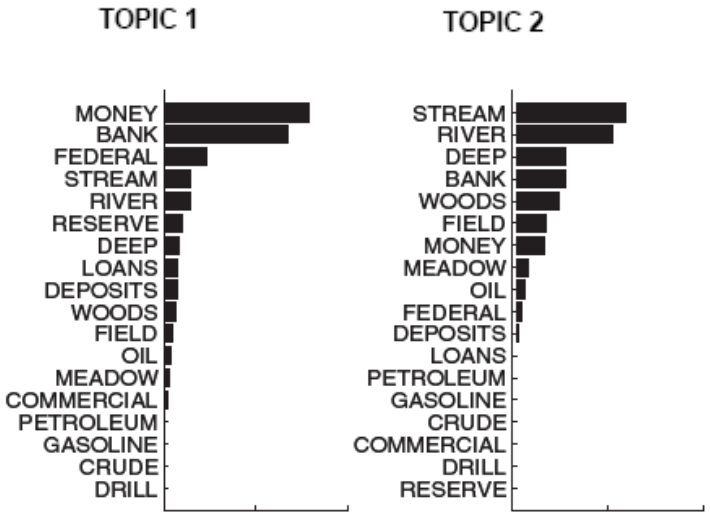
(in contrary to vector(Top2Vec)/transformer(BERTopic))

Documents are generated using a statistical generative process, such that each document is a mixture of topics, and each topic is a distribution over words. (Dirichlet distributions)

The goal of LDA is to infer the mixture of topics that a document describes from the words it contains.

It maps each document in a corpus to a set of topics by assigning topics to arrangements of words. If new documents are added, it can determine topics for this as well.

LDA – more technically explained

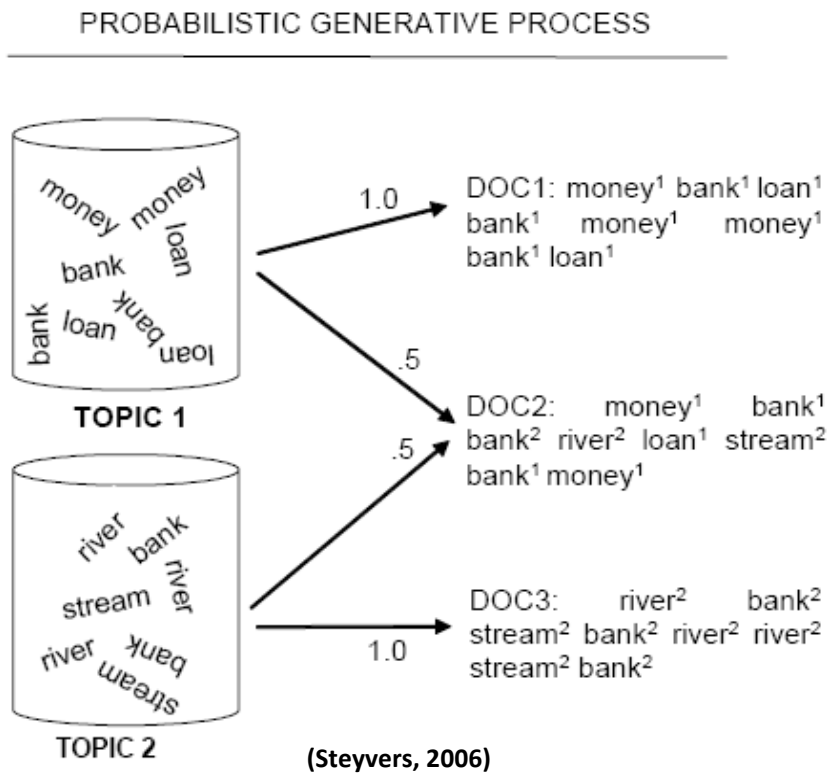


3 latent variables:

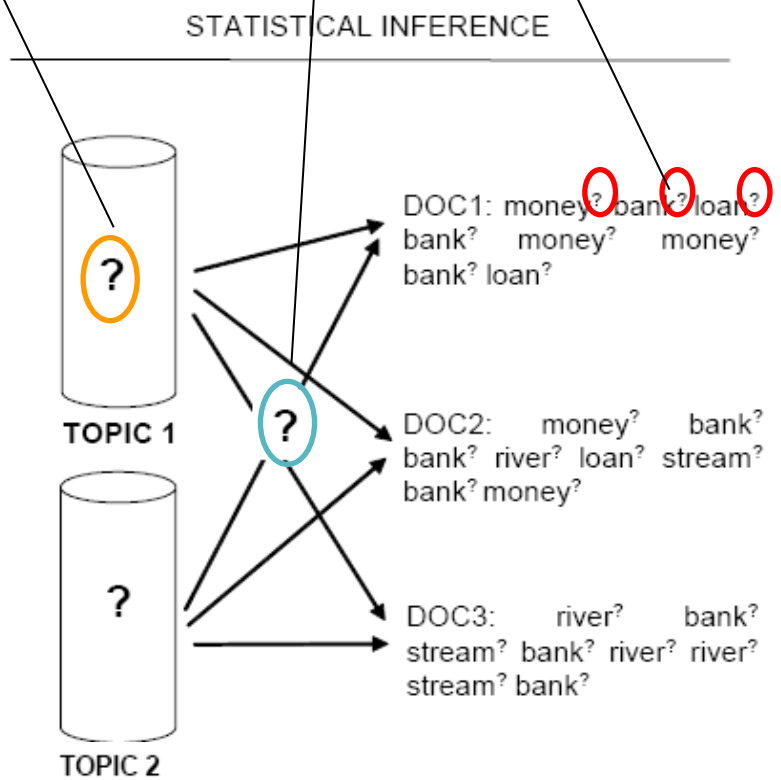
Word distribution per topic (word-topic-matrix) (β_k)

Topic distribution per doc (Topic proportion per doc) (topic-doc-matrix) (θ_d)

Topic word assignment ($Z_{d,n}$)



(Steyvers, 2006)



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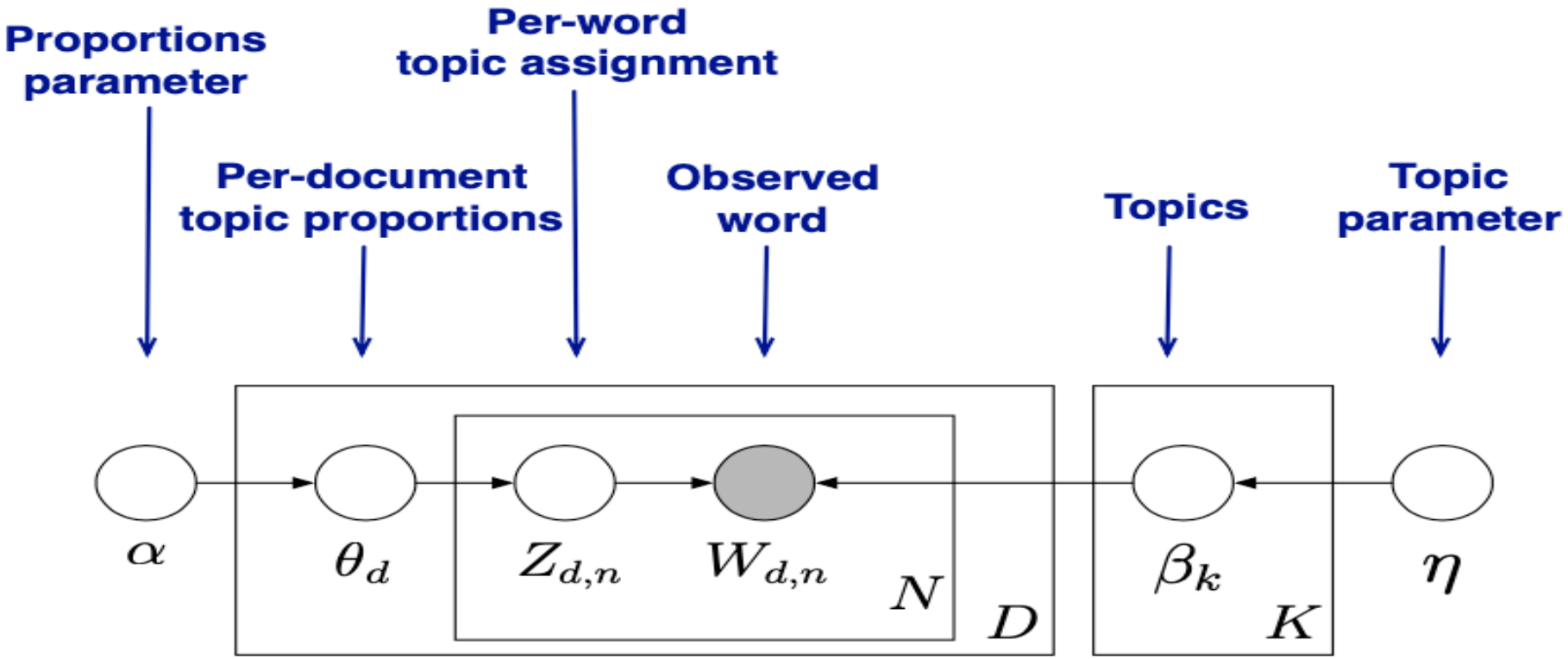
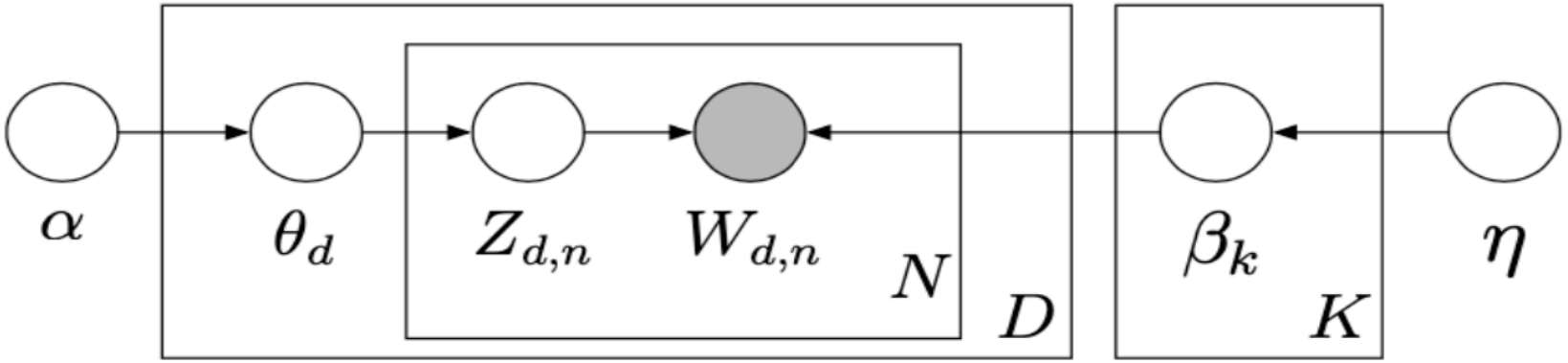
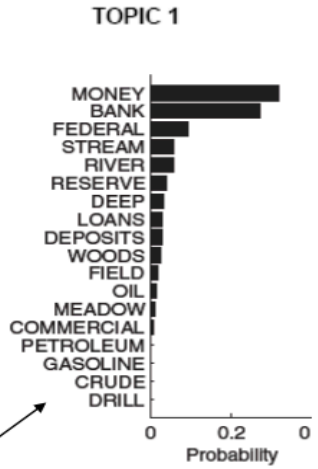
LDA as a graphical model

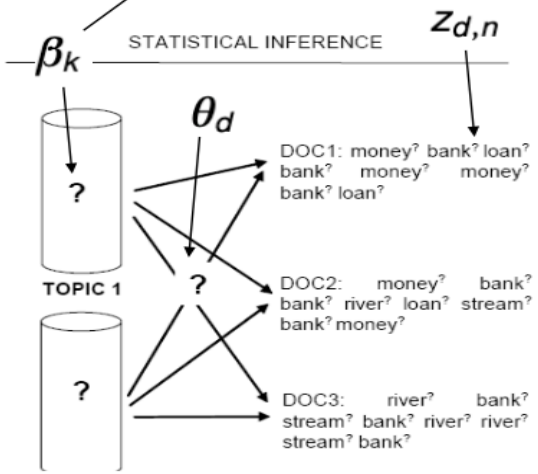


- Nodes are random variables; edges indicate dependence.
- Shaded nodes are observed; unshaded nodes are hidden.
- Plates indicate replicated variables.

LDA as a graphical model







- This joint defines a posterior, $p(\theta, z, \beta \mid w)$.
- From a collection of documents, infer
  - Per-word topic assignment $z_{d,n}$
  - Per-document topic proportions θ_d
  - Per-corpus Topic word distributions β_k
- Then use posterior expectations to perform the task at hand: information retrieval, document similarity, exploration, and others.

Why does LDA work?

- LDA trades off two goals.
  1. For each document, allocate its words to as few topics as possible.
  2. For each topic, assign high probability to as few terms as possible.
- These goals are at odds.
  - Putting a document in a single topic makes #2 hard:
    All of its words must have probability under that topic.
  - Putting very few words in each topic makes #1 hard:
    To cover a document's words, it must assign many topics to it.
- Trading off these goals finds groups of tightly co-occurring words.

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PARAMETER ESTIMATION PROBLEM

Latent Dirichlet Allocation

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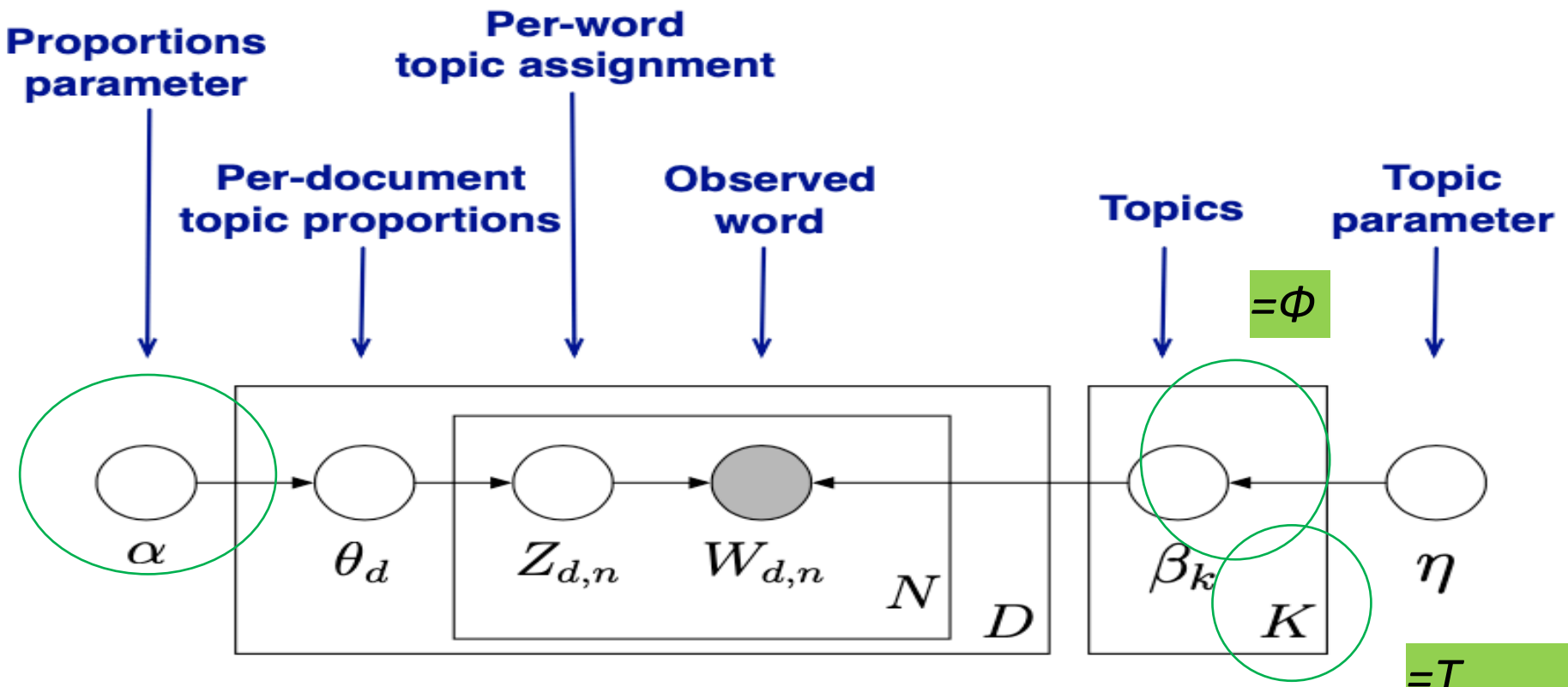
Hyperparameters in LDA

Document density factor (**α**)	Topic word density factor (**β**)	Number of topics selected (**K**)
• α controls the number of topics expected in the document. • Python ~ `passes`	• β controls the distribution of words per topic in the document	• K defines how many topics we need to assume/ distinquish • Python ~ `num_topics`

LDA as a graphical model



=T

(in Darling 2011)

- Nodes are random variables; edges indicate dependence.
- Shaded nodes are observed; unshaded nodes are hidden.
- Plates indicate replicated variables.

Bayes Theorem: P(hypothesis|oberservations)

Probabilistic model to describe the relationship between data (D) and a hypothesis (h):

”Posterier” that will be updated

”Prior” that needs to be estimated

$$P(h|D) = \frac{P(D|h) * P(h)}{P(D)}$$

Slides adapted from Jure Leskovec

Statistical Inference and Parameter Estimation

Key problem:

Compute posterior distribution of the hidden variables given a document

$$p(\underbrace{\theta, \mathbf{z}}_{\text{Latent Vars}} \mid \underbrace{\mathbf{w}, \alpha, \beta}_{\text{Observed Vars And Priors}}) = \frac{p(\theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta)}{p(\mathbf{w} \mid \alpha, \beta)}$$

Posterior distribution is intractable for exact inference.

Statistical Inference and Parameter Estimation

How can we estimate posterior distribution of hidden variables given a corpus of trainings-documents? (in a tractable way)

Direct	Indirect	Gibbs sampling
• Via expectation maximization, variational inference or expectation propagation algorithms	• Estimate the posterior distribution over z (i.e. P(z)) Topic assignment per word in doc: $P(z\|w) = Z_{d,n}$	• A form of Markov chain Monte Carlo, is often used to estimate the **posterior probability over a high-dimensional random variable, z**

Slides adapted from Jure Leskovec

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GIBBS SAMPLING

In Latent Dirichlet Allocation

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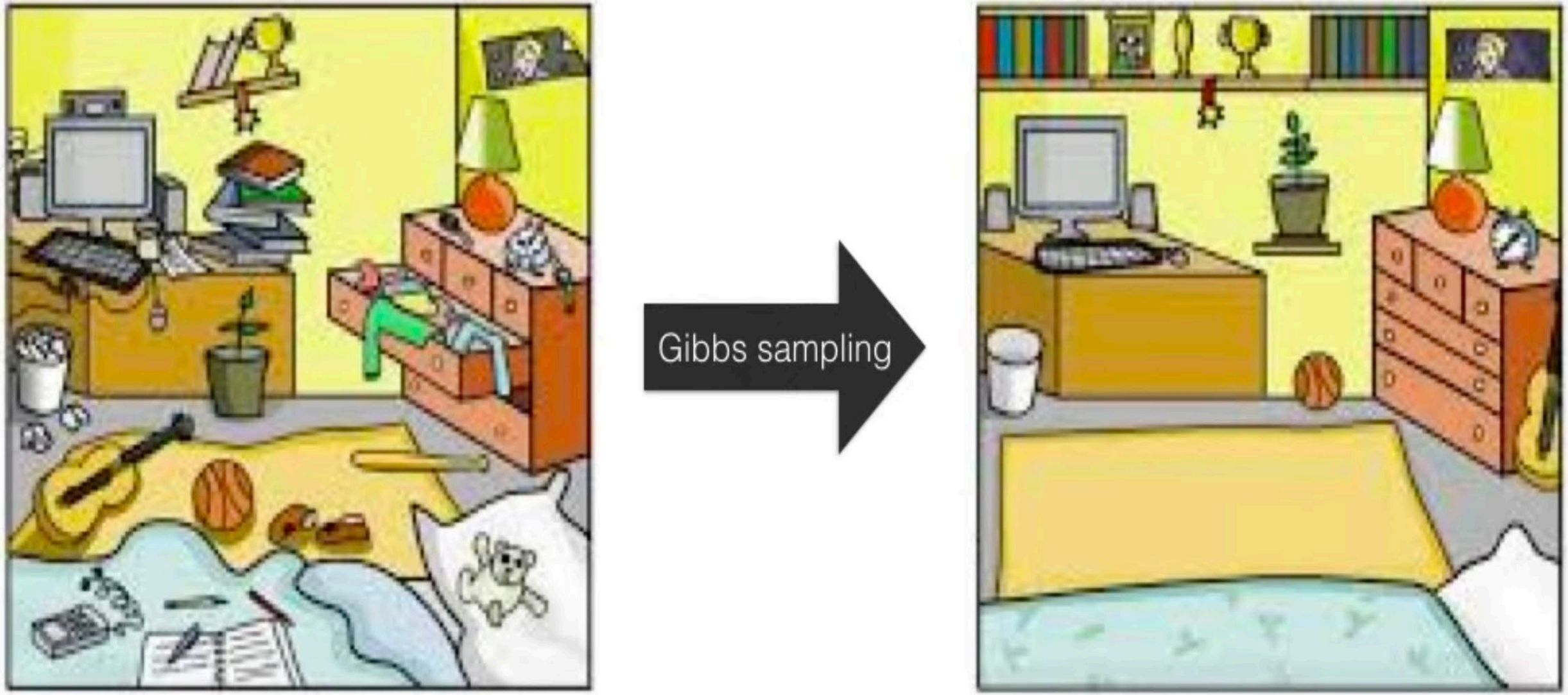
Gibbs Sampling



And what does Gibbs sampling do here?

A. Generates a sequence of samples from the joint probability distribution of **k (or T)** random variables.
B. It does so by computing posterior distribution over latent variable **z**
C. This is an iterative process and leads to the trade off between:
    1. For each document – allocate its words to as few topics as possible (topic proportion)
    2. For each topic, assign high probability to as few words as possible

Gibbs Sampling for LDA

Probability that topic ***j*** is chosen for word $\boldsymbol{w_i}$, conditioned on all other assigned topics of words in this doc and all other observed vars.

Count number of times a word token $\boldsymbol{w_i}$ was assigned to a topic ***j*** across all docs

$$P(z_i = j \mid \mathbf{z}_{-i}, w_i, d_i, \cdot) \propto \frac{\boxed{C^{WT}_{w_i j}} + \beta}{\sum_{w=1}^{W} \boxed{C^{WT}_{wj}} + W\beta} \frac{\boxed{C^{DT}_{d_i j}} + \alpha}{\sum_{t=1}^{T} \boxed{C^{DT}_{d_i t}} + T\alpha}$$

unnormalized!

Count number of times a topic ***j*** was already assigned to some word token in doc $\boldsymbol{d_i}$

=> divide the probability of assigning topic j to word wi by the sum over all topics T

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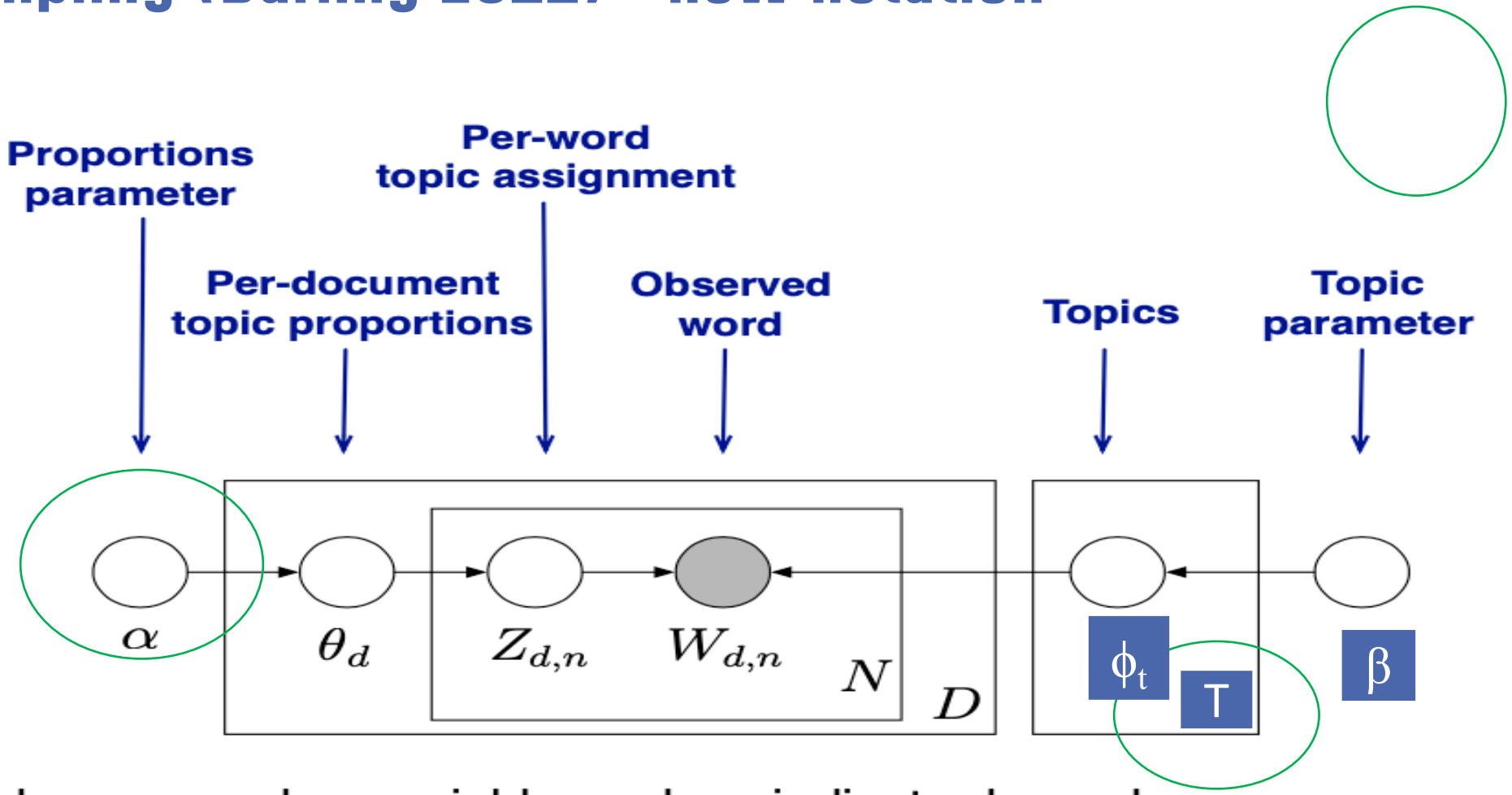
Gibbs Sampling (Darling 2011) – new notation



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Terms Darling 2011

Hidden:

- ϕ_k = Word distribution per topic in corpus, P(w, z) (sometimes called β_k)
- Θ_d = Topic distribution per document, P(z)
- $Z_{d,n}$ = Topic assignment per word in doc: P(z|w)

Not hidden

- w = Observed word distribution in all words, W (sometimes called N)
- C^{WT} = Count word token **wi** per topic **j**
- C^{DT} = Count topic **j** already assigned per doc **di**
- D = corpus of documents (of M documents)
- T = all topics (sometime called K)
- α = topics per collection, hyperparameter
- β = Hyper parameter (sometimes called η)

Intuition: We update the P(z|w) based on what we know, including all the other words' topics, z_{-i}

$$P(z_i = j \mid \mathbf{z}_{-i}, w_i, d_i, \cdot) \propto \frac{C^{WT}_{w_i j} + \beta}{\sum_{w=1}^{W} C^{WT}_{wj} + W\beta} \frac{C^{DT}_{d_i j} + \alpha}{\sum_{t=1}^{T} C^{DT}_{d_i t} + T\alpha}$$

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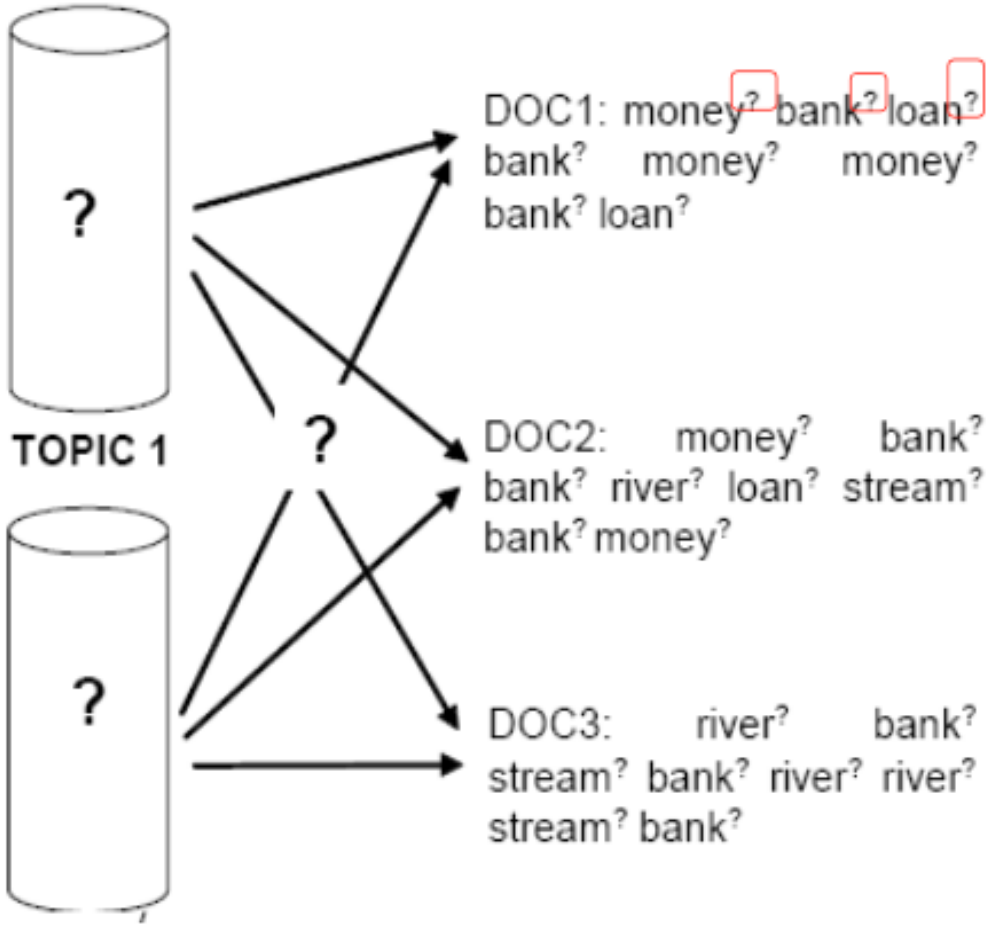
Gibbs Sampling for LDA



Random start

Iterative

For each word we compute

$$P(z_i = j \mid \mathbf{z}_{-i}, w_i, d_i, \cdot) \propto \frac{C^{WT}_{w_i j} + \beta}{\sum_{w=1}^{W} C^{WT}_{wj} + W\beta} \frac{C^{DT}_{d_i j} + \alpha}{\sum_{t=1}^{T} C^{DT}_{d_i t} + T\alpha}$$

How dominant is a topic z in the doc d?

How often was the topic z already used in doc d?

How likely is a word for a topic z?

How often was the word w already assigned to topic z?

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Run Gibbs Sampling procedure - algorithm

- Start: assign each word token to a random topic
- $\mathbf{C}^{WT}$ = Count number of times a word token ***wi*** was assigned to a topic ***j***
- $\mathbf{C}^{DT}$ = Count number of times a topic ***j*** was already assigned to some word token in doc ***di***
- First Iteration:
  - For each word token, the count matrices $\mathbf{C}^{WT}$ and $\mathbf{C}^{DT}$ are first decremented by one for the entries that correspond to the current topic assignment
  - Then, a new topic is sampled from the current topic-distribution of a doc and the count matrices $\mathbf{C}^{WT}$ and $\mathbf{C}^{DT}$ are incremented with the new topic assignment.
- Each Gibbs sample consists the set of topic assignments to all *N* word tokens in the corpus, achieved by a single pass through all documents

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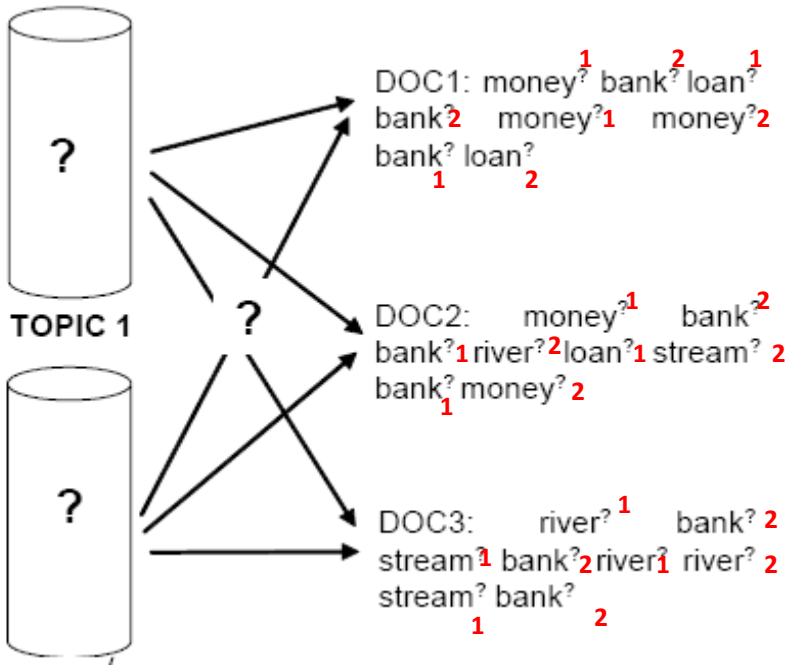

Run Gibbs Sampling Example (0)

1. Random topic assignments
2. 2 count-matrices:

C^{WT} → Words per topic

C^{DT} → Topics per document

	topic1	topic2
money	3	2
bank	3	6
Loan	2	1
River	2	2
Stream	2	1

	doc1	doc2	doc3
topic1	4	4	4
topic2	4	4	4

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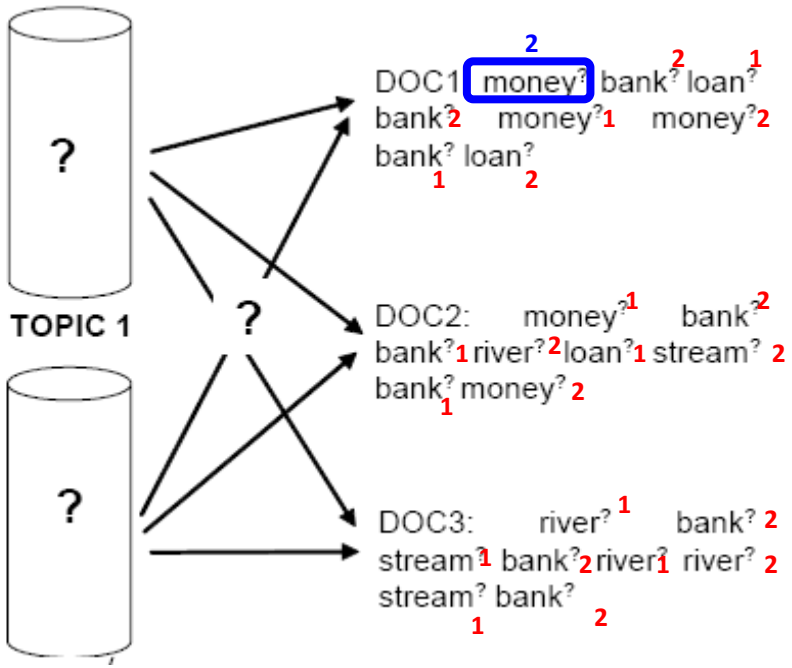

Run Gibbs Sampling Example (1)

First Iteration:

- Decrement $\mathbf{C}^{DT}$ and $\mathbf{C}^{WT}$ for current topic j
- Sample new topic from the current topic-distribution of a doc

	topic1	topic2
money	3 2	2
bank	3	6
Loan	2	1
River	2	2
Stream	2	1

	doc1	doc2	doc3
topic1	4 3	4	4
topic2	4	4	4

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Run Gibbs Sampling Example (1) - calculating

α = 50/T = 25 and β = 0.01

Count number of times a word token $\boldsymbol{w_i}$ was assigned to a topic $\boldsymbol{j}$ across all docs

How often was topic j used in doc $\boldsymbol{d_i}$

$$P(z_i = j \mid \mathbf{z}_{-i}, w_i, d_i, \cdot) \propto \frac{C^{WT}_{w_i j} + \beta}{\sum_{w=1}^{W} C^{WT}_{wj} + W\beta} \frac{C^{DT}_{d_i j} + \alpha}{\sum_{t=1}^{T} C^{DT}_{d_i t} + T\alpha}$$

Unique words, **W** in doc

How often were all topics used in doc $\boldsymbol{d_i}$

From Doc 1:

money=1 **Money=2**

	topic1	topic2
money	3 2	2
bank	3	6
Loan	2	1
River	2	2
Stream	2	1

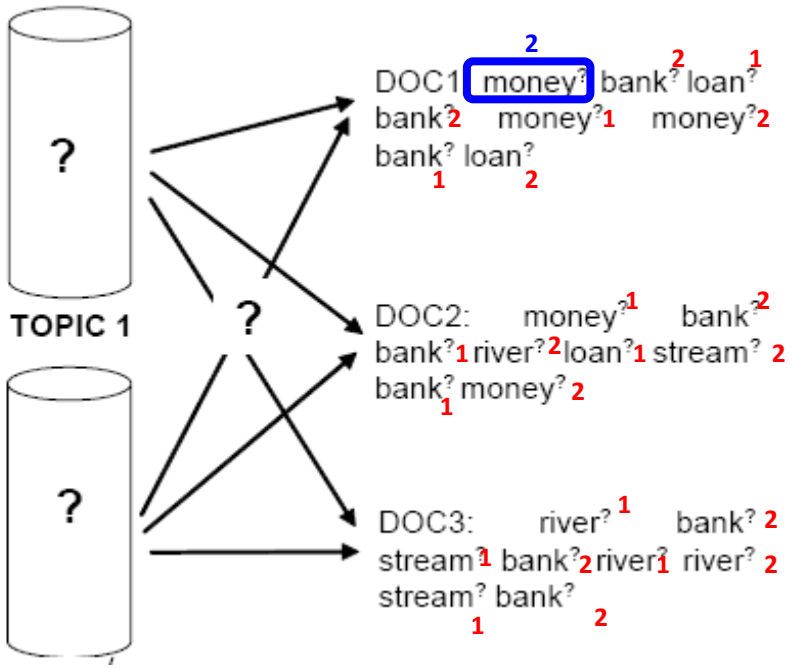
$$P(z_i = topic1 | z_{-i}, \boldsymbol{money}, d_i, .) = \frac{2 + 0.01}{11 + 5 * 0.01} * \frac{3 + 25}{7 + 2 * 25} = 0.089$$

$$P(z_i = topic2 | z_{-i}, \boldsymbol{money}, d_i, .) = \frac{2 + 0.01}{12 + 5 * 0.01} * \frac{4 + 25}{7 + 2 * 25} = 0.085$$

	doc1	doc2	doc3
topic1	4 3	4	4
topic2	4	4	4

"Money" is sampled by the two based on the probability-weights and assigned to Topic 2 (it could also have been Topic 1). Next time Money appears in doc1 it will be topic 2 with even higher probability

Run Gibbs Sampling Example (1)



First Iteration:

- Decrement $\mathbf{C}^{DT}$ and $\mathbf{C}^{WT}$ for current topic j
- Sample new topic from the current topic-distribution of a doc
- **Tables are now updated**

	topic1	topic2
money	2	3
bank	3	6
Loan	2	1
River	2	2
Stream	2	1

	doc1	doc2	doc3
topic1	3	4	4
topic2	5	4	4

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INFERENCES OF LDA

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Practical requirements of LDA

- Main method for the past 20 years
- Relies on Bag-of-words (ignoring semantic context)
- Relies on a nice cleaning (stop words, stemming, lemmatization, etc.)
- Requires that you have an idea about the number of topics in advance
- Does not work well in practice on small ”documents” (short tweets, headers, etc)

A bit like k-means clustering

Example inference: Topics in a document

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

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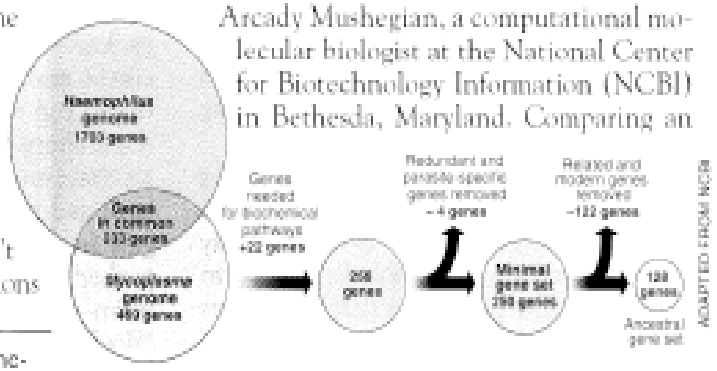

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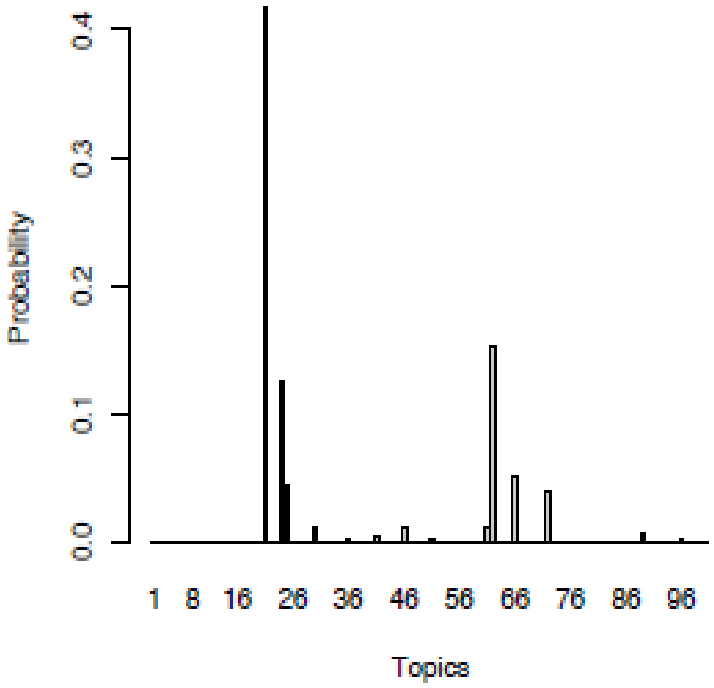

Topics vs. words – note that there are no "names"

How can these be named?

human	evolution	disease	computer
genome	evolutionary	host	models
dna	species	bacteria	information
genetic	organisms	diseases	data
genes	life	resistance	computers
sequence	origin	bacterial	system
gene	biology	new	network
molecular	groups	strains	systems
sequencing	phylogenetic	control	model
map	living	infectious	parallel
information	diversity	malaria	methods
genetics	group	parasite	networks
mapping	new	parasites	software
project	two	united	new
sequences	common	tuberculosis	simulations

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Visualizing a document

Abstract with the most likely topic assignments

Statistical approaches help in the determination of significant configurations in protein and nucleic acid sequence data. Three recent statistical methods are discussed: (i) score-based sequence analysis that provides a means for characterizing anomalies in local sequence text and for evaluating sequence comparisons; (ii) quantile distributions of amino acid usage that reveal general compositional biases in proteins and evolutionary relations; and (iii) *r*-scan statistics that can be applied to the analysis of spacings of sequence markers.

- Use the posterior topic probabilities of each document and the posterior topic assignments to each word

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TOP2VEC

Topic modelling using vector semantics

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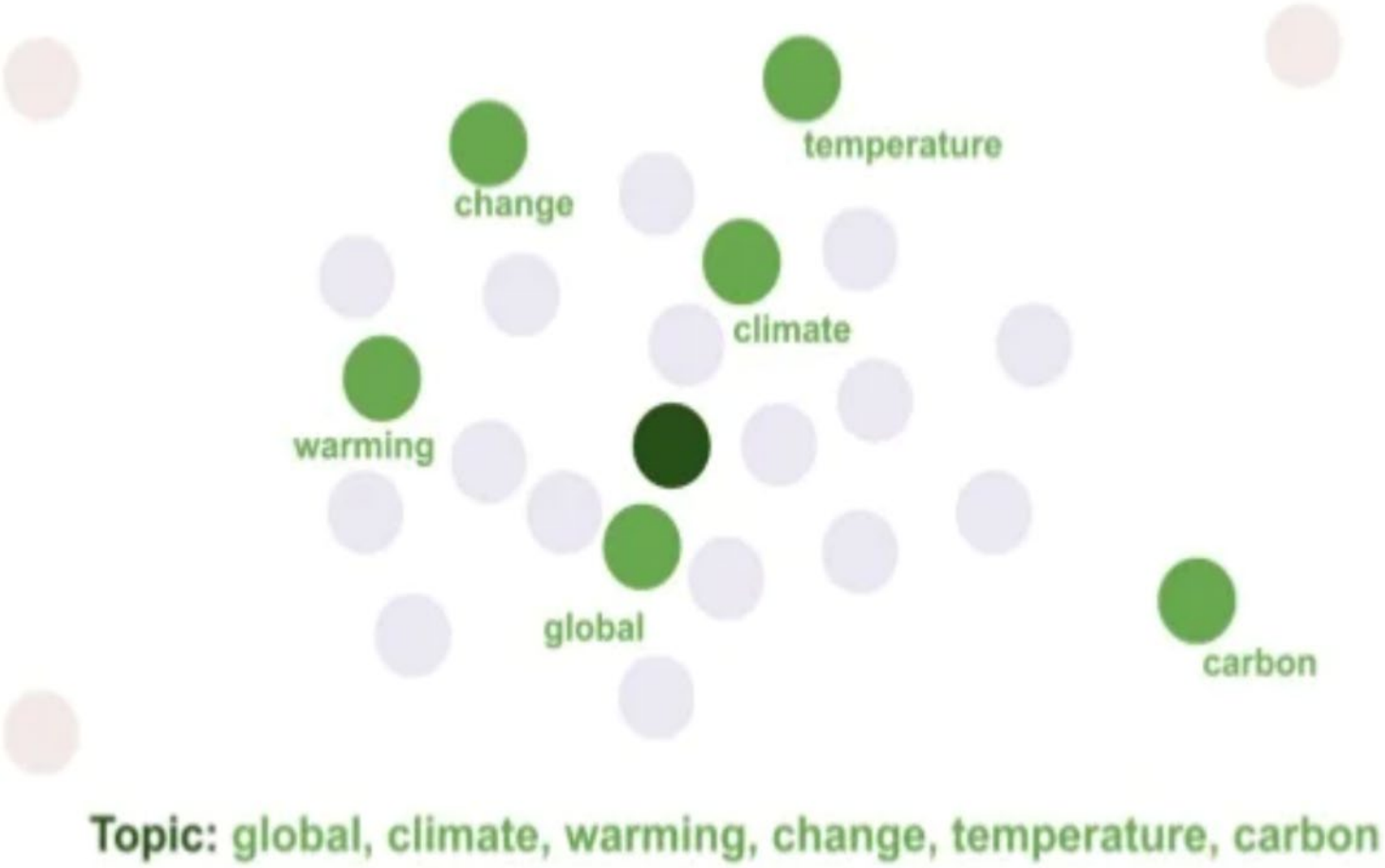
Top2Vec: Distributed Representation of Topics

Recent arxiv paper (2020)

GitHub: ddangelov/Top2Vec – and more…

Does not require the number of topics to be known in advance

Does not require pre-processing



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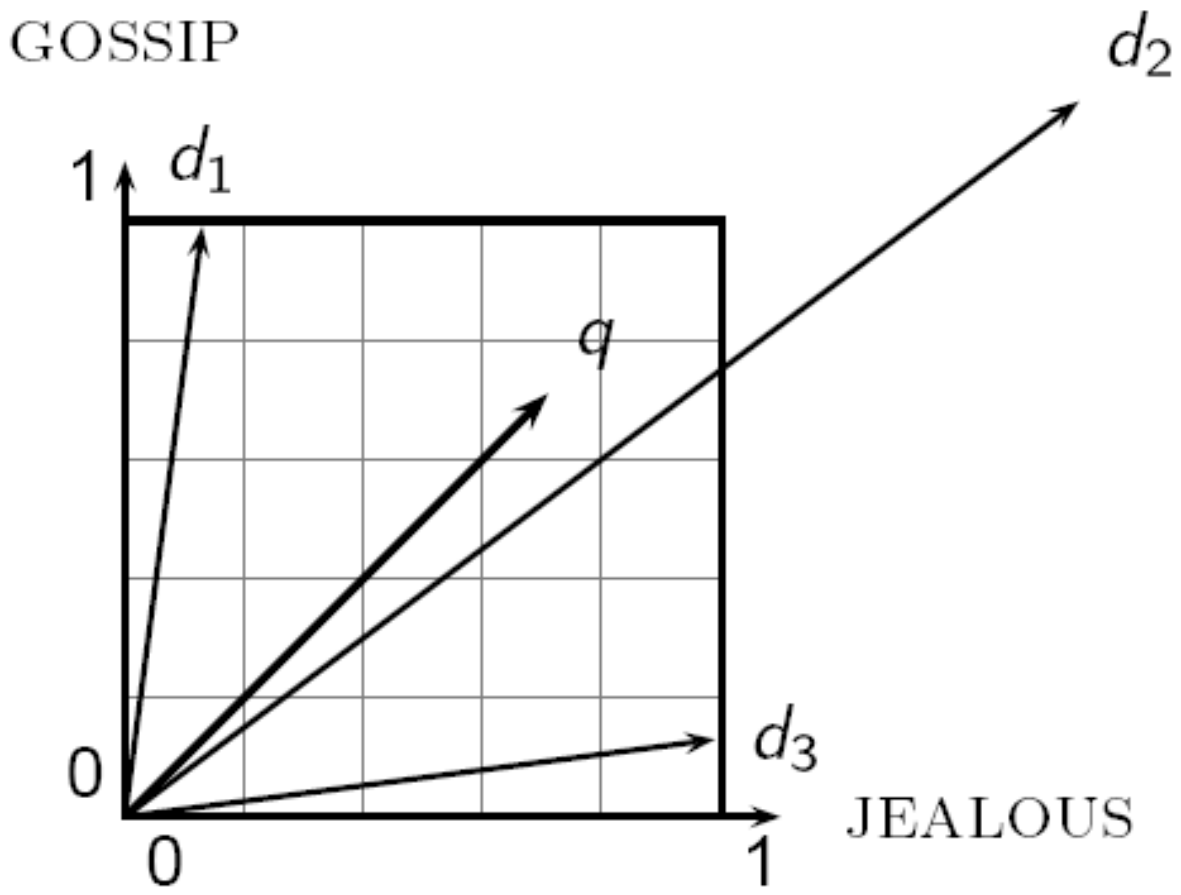
You will learn more on word embedding next week, but here is an intuition

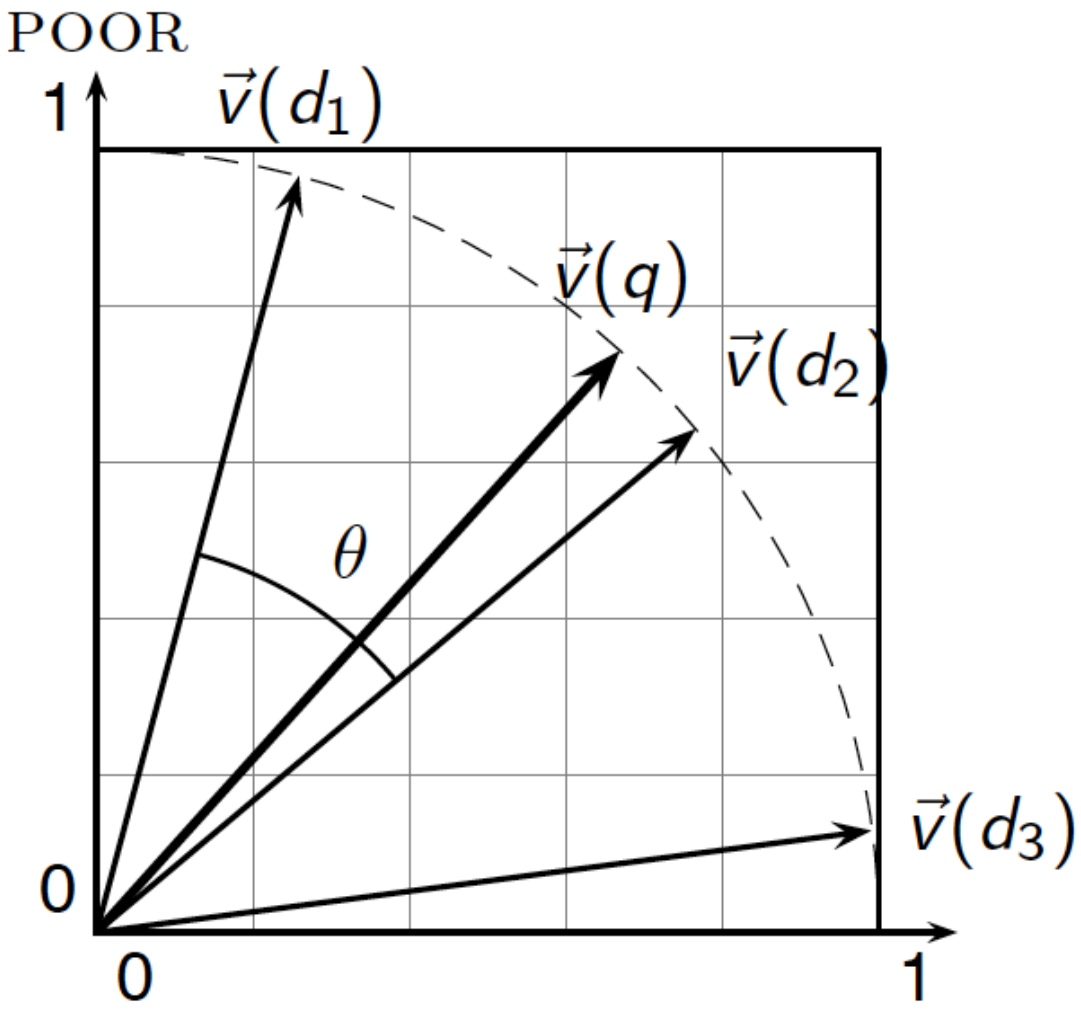
So we have a vector space cor. number of words

- Terms are axes of the space
- Documents are points or vectors in this space
- d=word or document



Slides adapted from Jure Leskovec

UMAP use Cosine similarity (smallest angles in n-dimensional room)

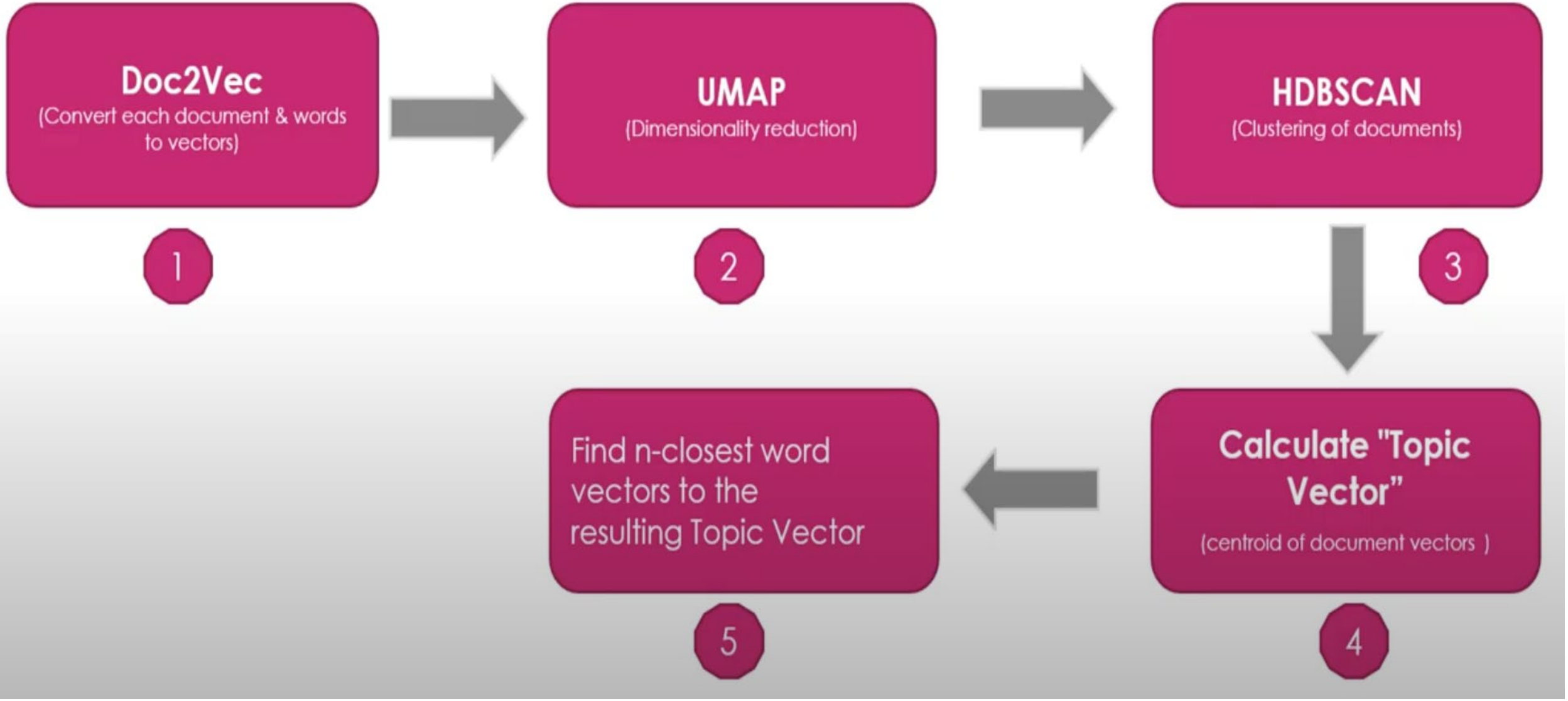


2-dimensional example

(from another application d=documents or words)

N can be 100-millions

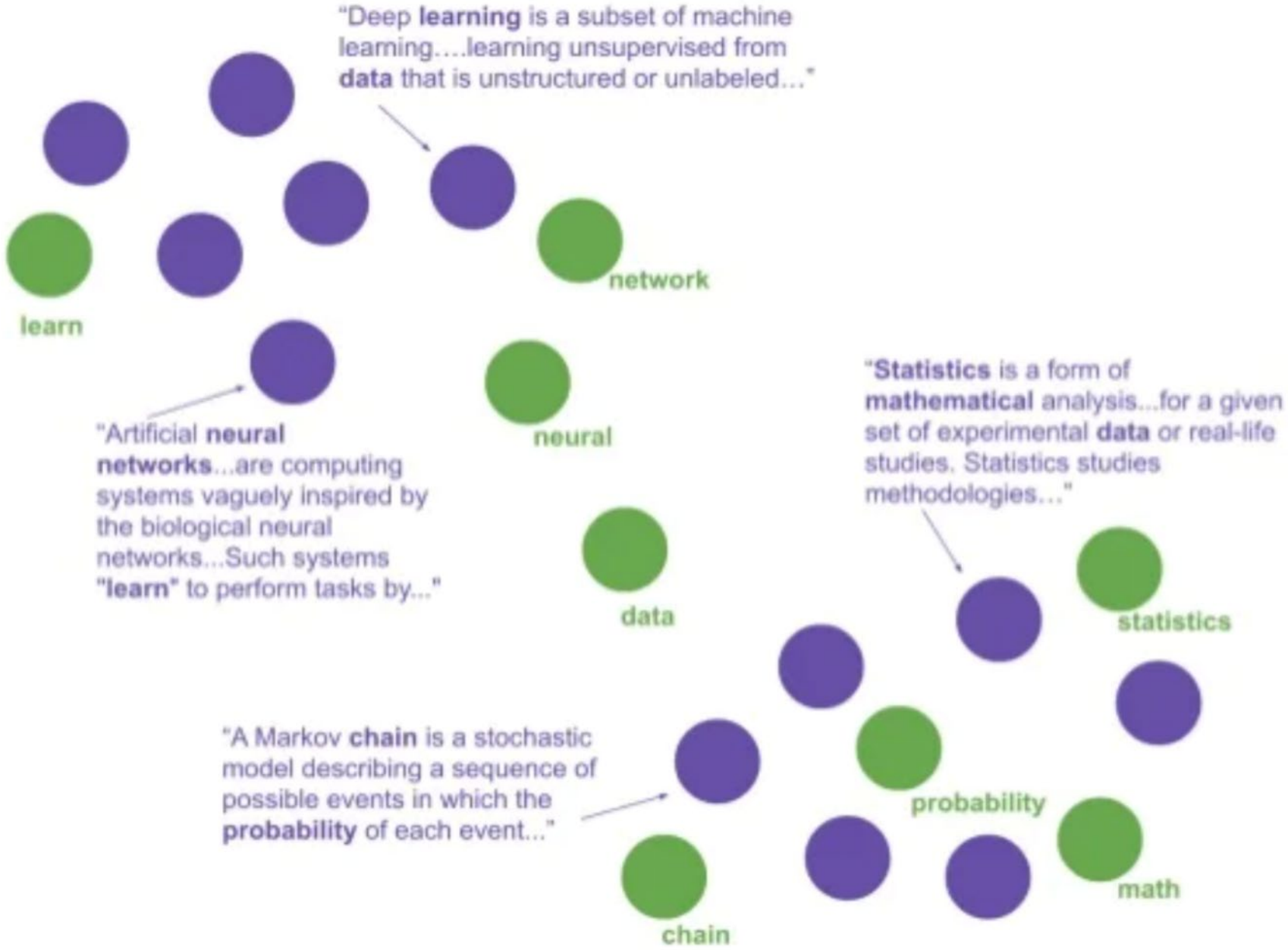
Top2Vec algorithm: How does it work?



How does it work? 5-step Algorithm

Step 1 doc2vec

- Create an embedding of each document
- For example, using:
  - Universal Sentence Encoder,
  - BERT Sentence Transformer



How does it work?

5-step Algorithm

Step 2: Dimensionality reduction

- Project the embedding to a smaller dimension (e.g., 2D)
- For example, using UMAP

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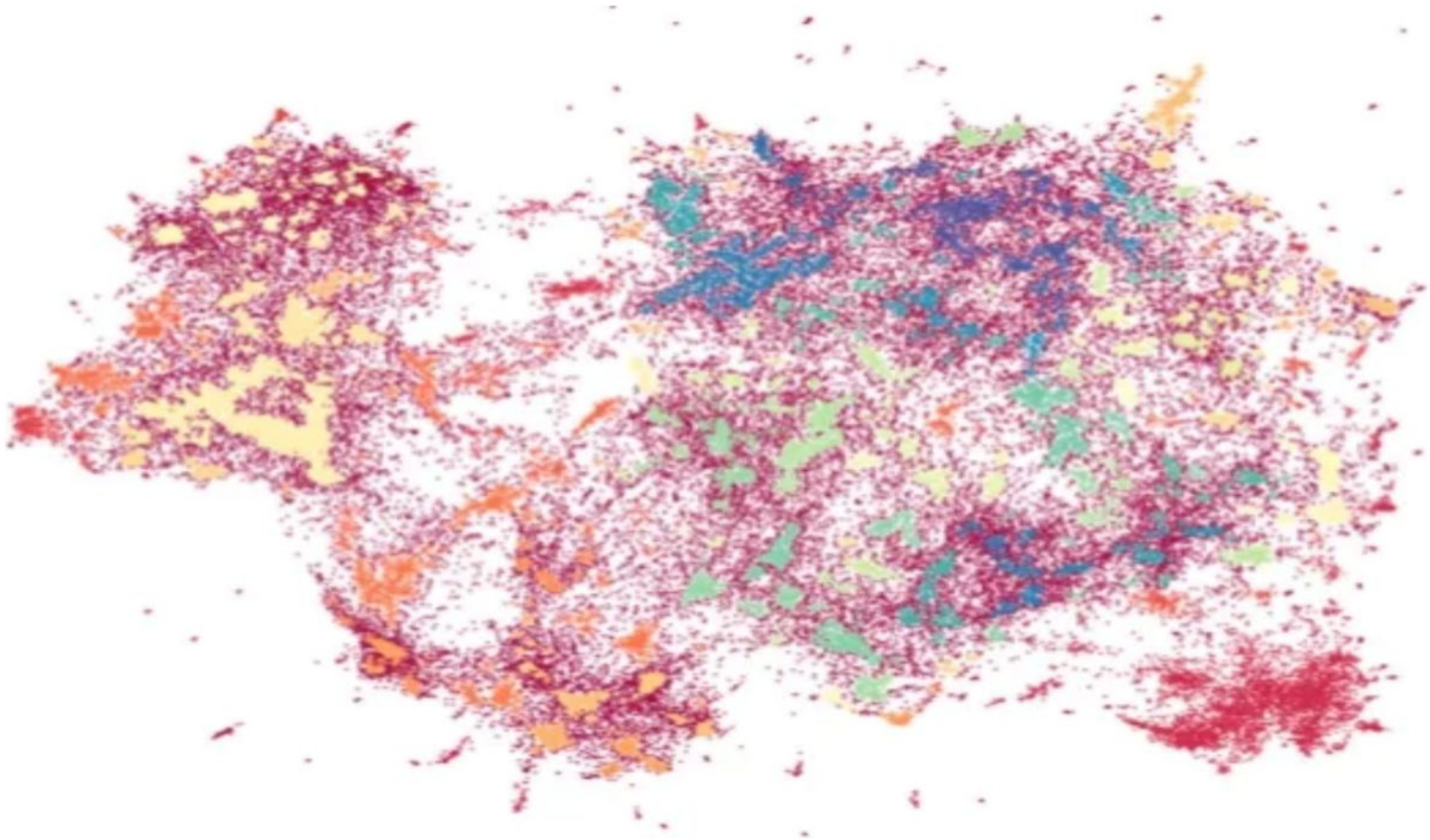
How does it work?

5-step Algorithm

Step 3: Clustering

- Find dense clusters of embeddings; a cluster is a topic
- For example, using: HDBSCAN



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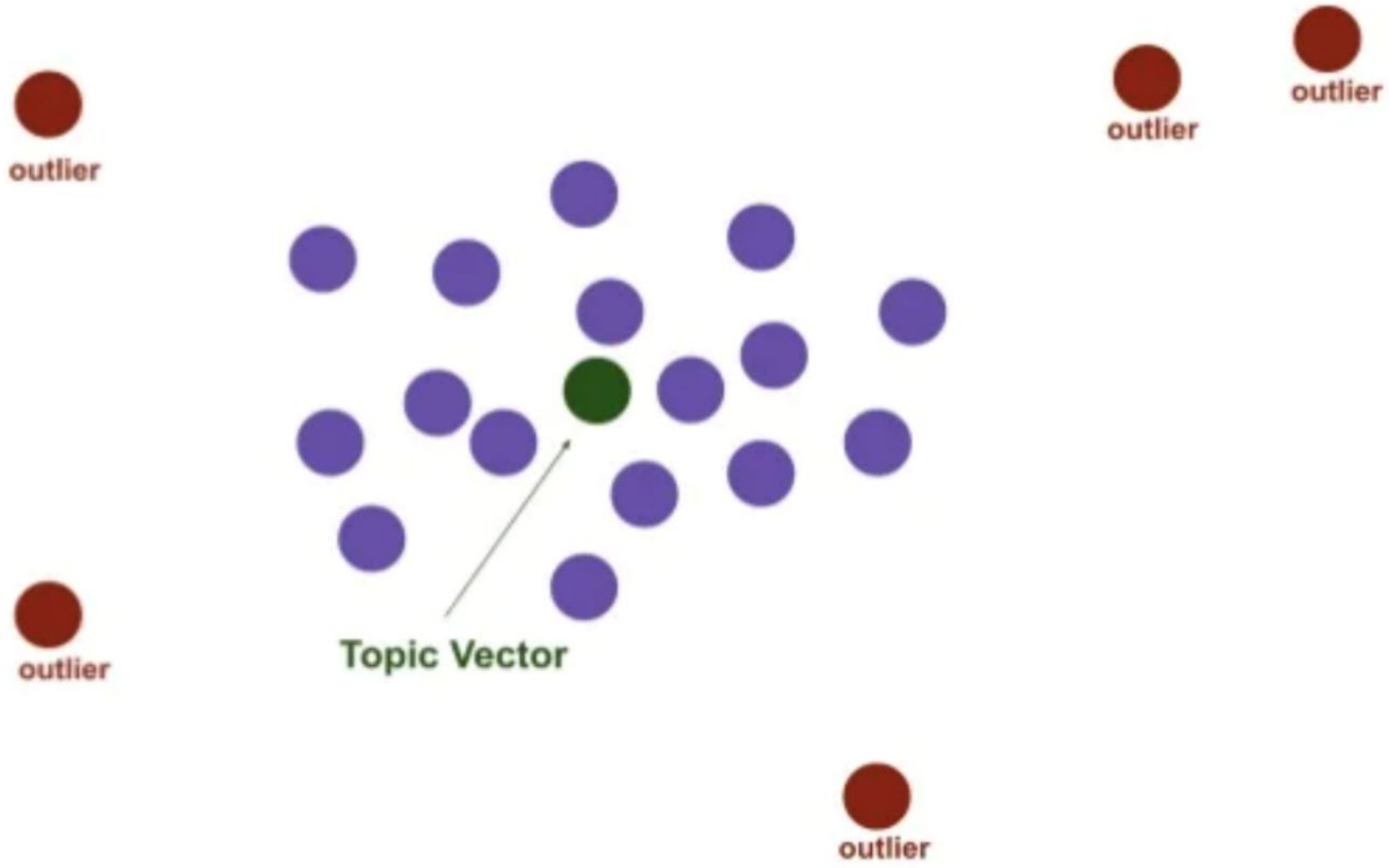
How does it work?

Step 4: Calculate topic vector

- Find the centroid of each embedding cluster in original high dimension
- This centroid is the topic vector



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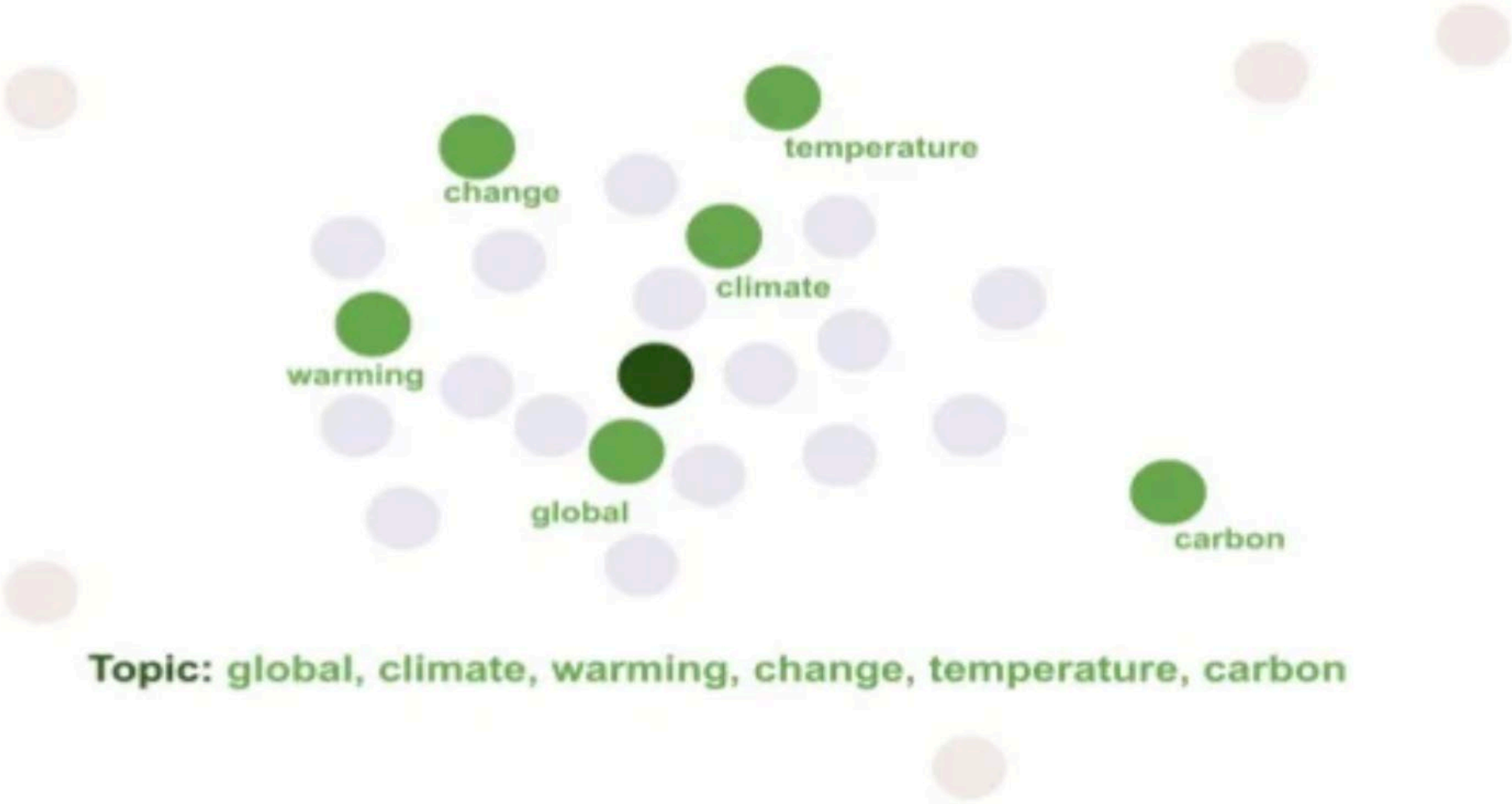
Top2Vec

Step 5

- Find the n-closest word vectors to the topic vector

- How?



These words define the topic

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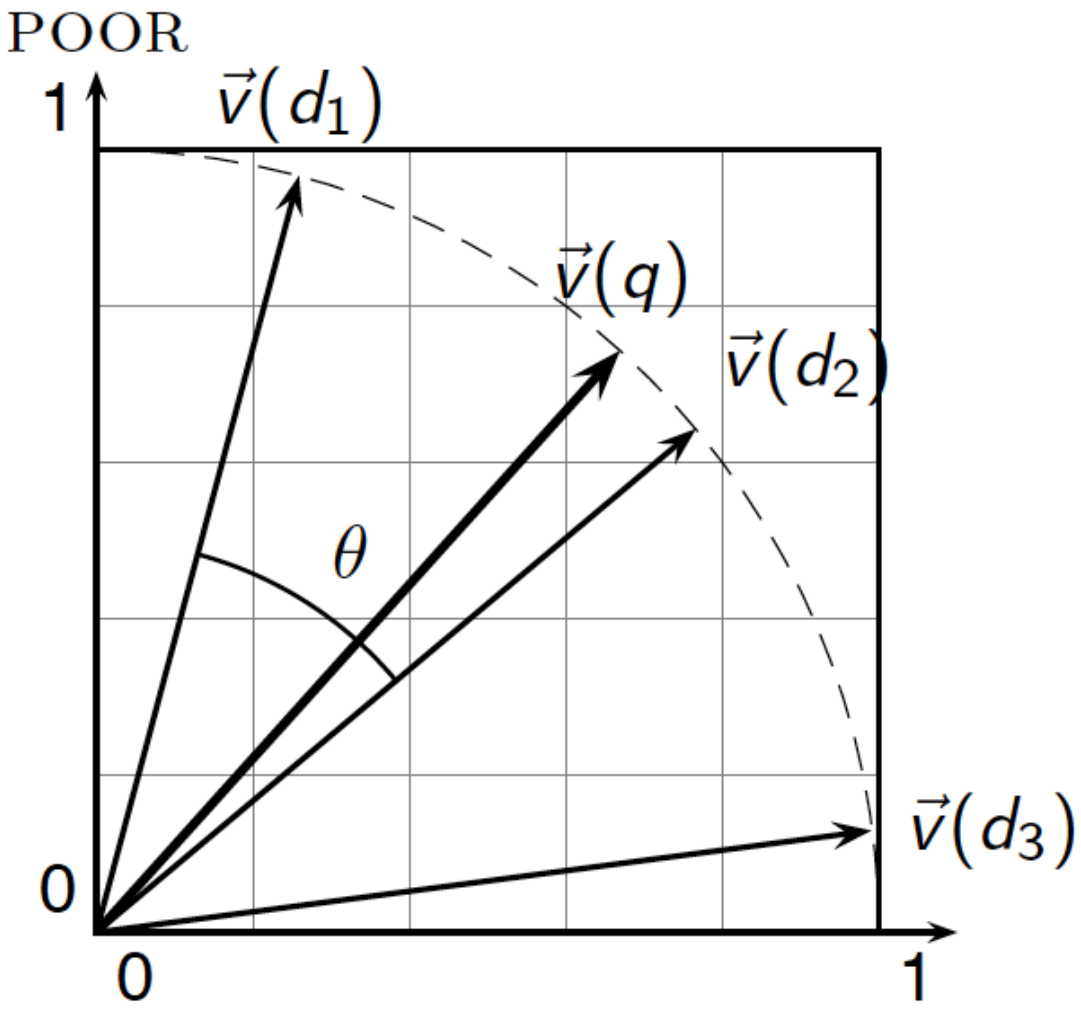
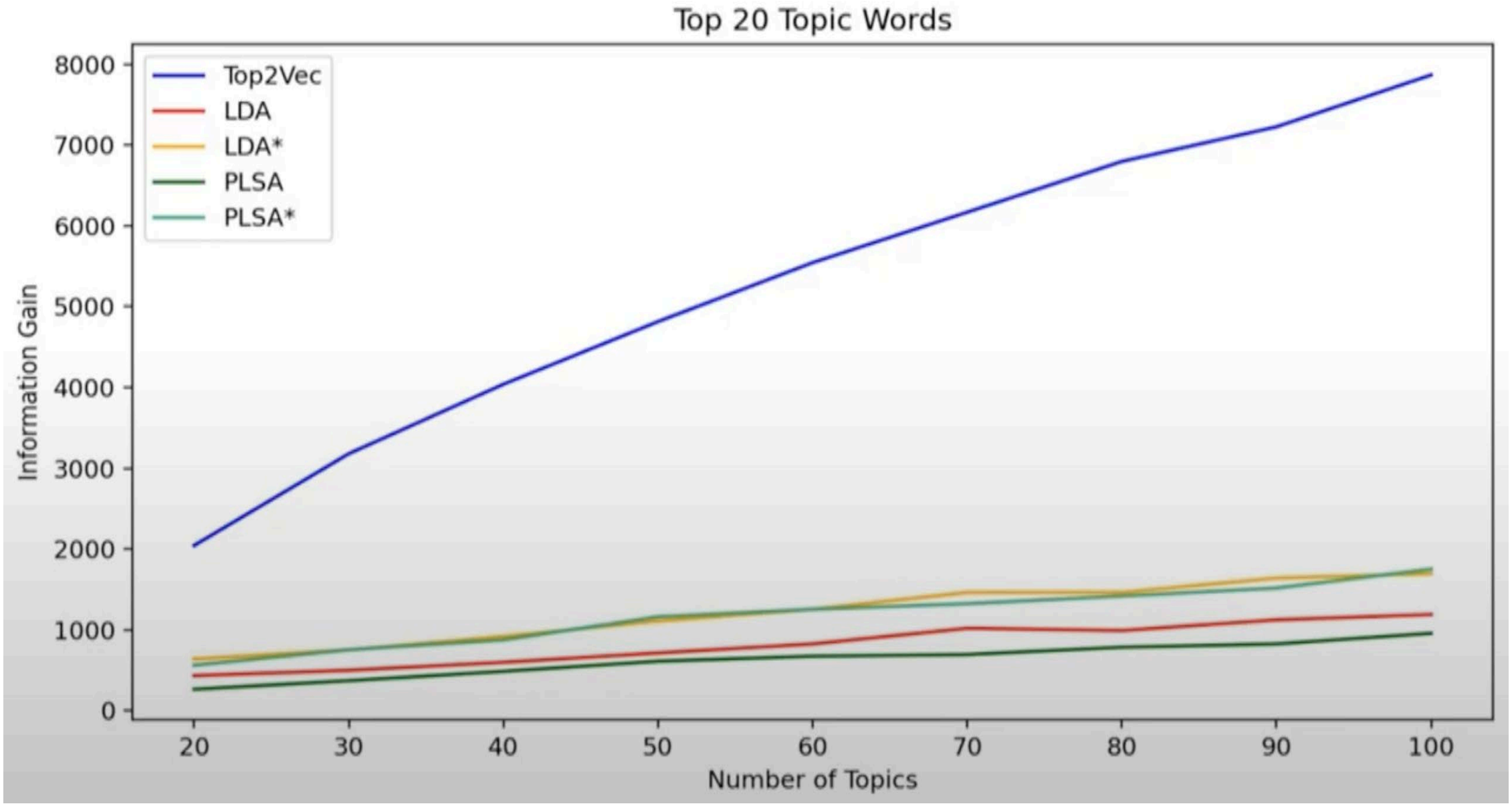
AND we again use Cosine similarity (smallest angles in n-dimensional room)



2-dimensional example

(from another application ”d” are now words)

- Performance – according to Angelov



Top2Vec: Many Hyperparameters

- Hyper-parameters Tuning
  - Doc2Vec : vector size = **300**, window =**15**
  - UMAP : Default is Nearest Neighbors = **15**, metric = "**cosine**"
  - HDBSCAN : min cluster size = **15**, metric = "**euclidean**"

Top2Vec: Pros

- ✓ **Find number of topics itself** (and thus does not loose semantics)
- ✓ Stop word removal, stemming/lemmatization is not required
- ✓ Better packages now

Top2Vec: Cons

- **Time consuming**
  - Use small dataset for initial testing
  - Get more resources in order to use big dataset

- Experience from exam reports – they are hard to tune correct (**many hyperparameters**)
- You need to experiment with different top2vec packages

Slides adapted from Jure Leskovec

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BERTOPIC

Topic modelling

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Attention is all you need 2017/2018

- **What is this?**
- TRANSFORMERS
- Semantic similarity by vectors (~top2vec)
- Contextual embedding based on architecture and pretraining (solves the ”bank”-river-finance problem)
- Computational heavy but high performance
- Other applications?

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

BERTtopic – a close cousing to Top2Vec

First product was BERT

BERTopic

- A bit like Top2Vec
- Does not calculate centroids: Consider all the documents in a cluster as unique documents
- Extracts topics using a class-based variation of TF-IDF (similar logic to LDA-tradeoff)
- As it is transformer-based it is even more costly

Trade offs LDA, Top2Vec, BERTopic: Three Generations of Topic Models

	LDA	Top2Vec	BERTopic
Type	Probabilistic	Embedding + clustering	Transformer + clustering
Representation	Bag-of-words	Doc2Vec embeddings	BERT embeddings
Semantics	Weak	Medium	Strong
Interpretability	High	Medium	High (usually)
Choose K yourself?	Yes ❗	No	No
Speed	Fast	Medium	Slow ❗
Scalability	High	Medium	Lower
Short texts	Poor ❗	Better	Best
Similarity among topics	JS divergence/euclidian	Cosine	Cosine

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FINISHING OFF

Topic modelling

Ole Torp Lassen
Teaching Associate Professor, DIGI

Sine Zambach
Assistant Professor, DIGI

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Recap – what is Topic modelling good for?

- Discover topics within a corpus (costumer reviews, interviews, news articles, tweets)
- Assign topics to a new document in the corpus (self-supervised learning!)
- Calculate similarity between documents
- …..

Slides adapted from Jure Leskovec

Extending topic modelling: Dynamic topic models

1789

My fellow citizens: I stand here today humbled by the task before us, grateful for the trust you have bestowed, mindful of the sacrifices borne by our ancestors...

Inaugural addresses

2009

AMONG the vicissitudes incident to life no event could have filled me with greater anxieties than that of which the notification was transmitted by your order...

- LDA assumes that the order of documents does not matter.
- Not appropriate for sequential corpora (e.g., that span hundreds of years)
- Further, we may want to track how language changes over time.
- Dynamic topic models let the topics *drift* in a sequence.

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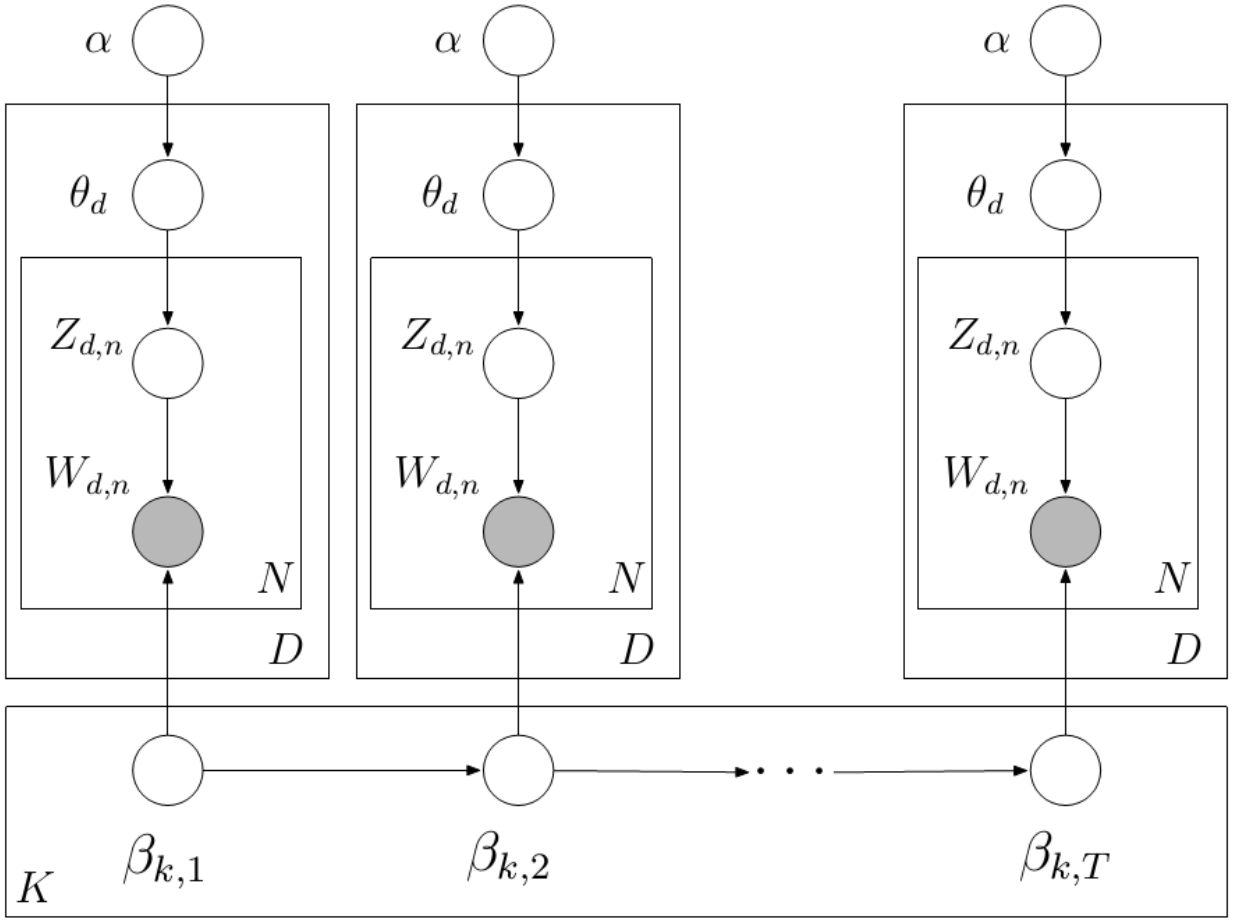
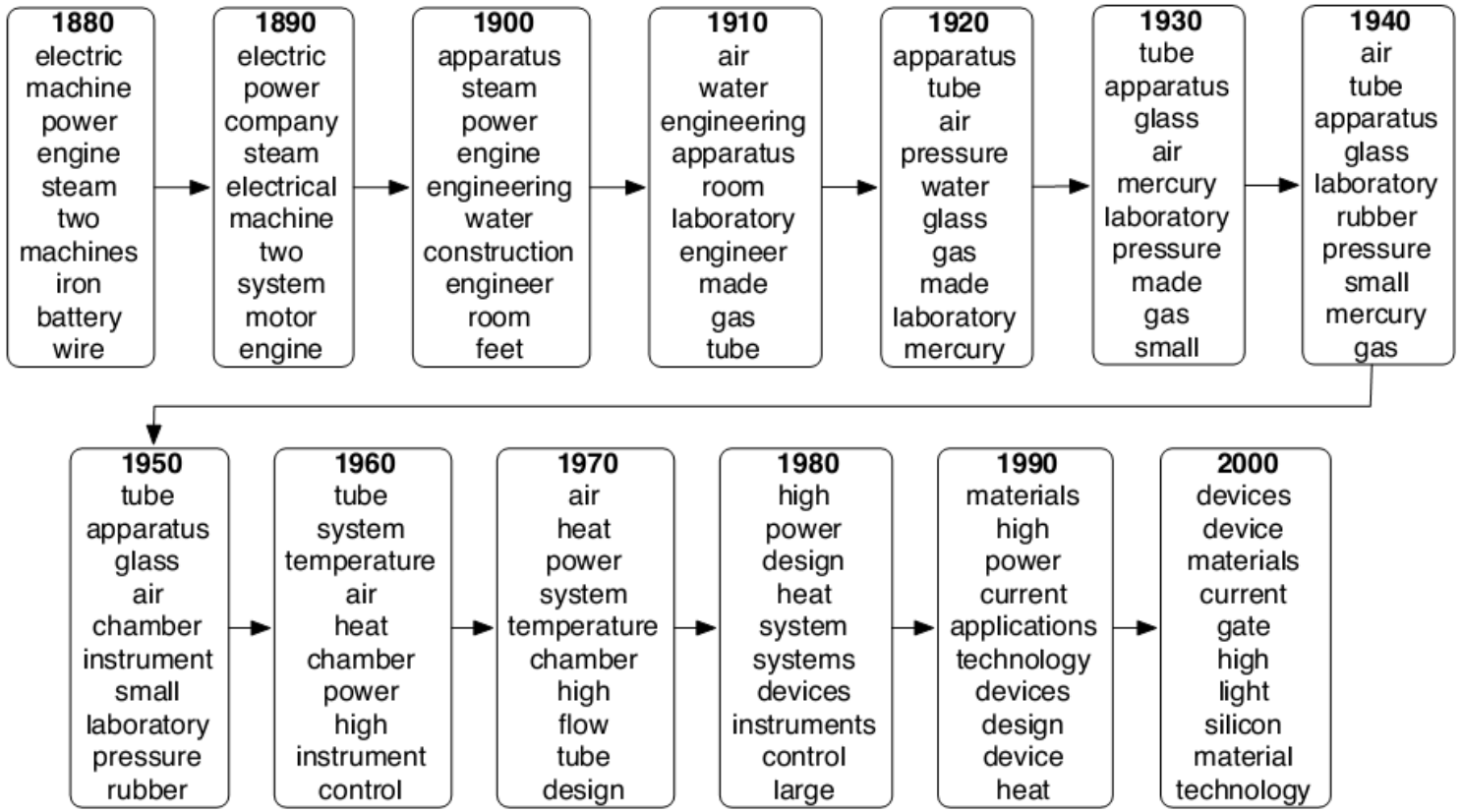
Dynamic topic models



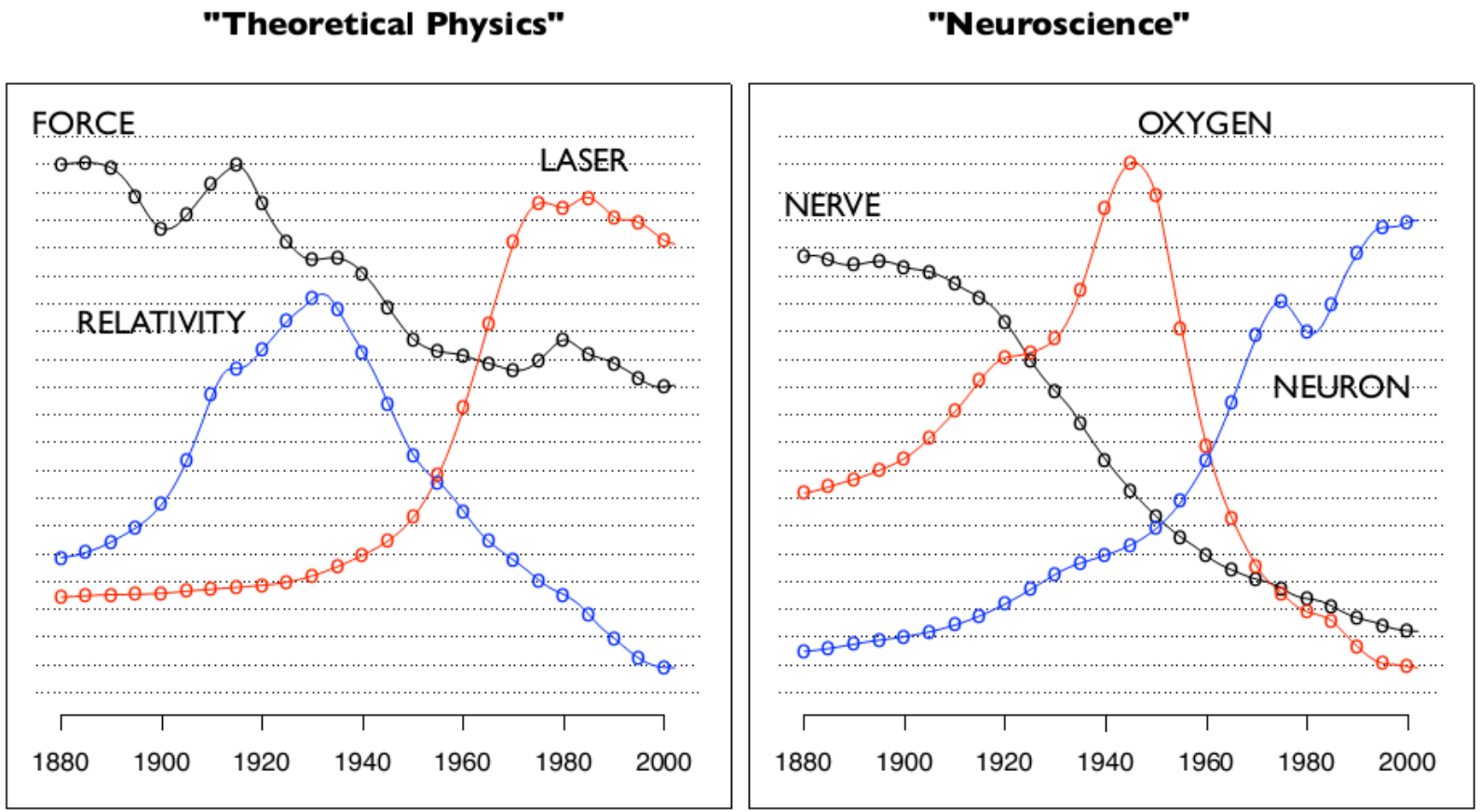
Topics drift through time

Dynamic topic models Technical reports



Dynamic topic models

(though we have names as Neuroscience for the topics)



What should you know from todays lecture?

- Understand – overall – what is topic modelling and how can it be used?
- Understand intuition behind LDA
  - How it generates topics
- Understand why and how we use gibbs sampling
  - That it helps with parameter estimation and how (but you dont need to be able to calculate a random new example)
- Understand intuition behind Top2Vec and BERTopic
  - How does this generates topics and what are the pros and cons compared to LDA?

Good References

- Blei and topic models (2 LONG, but quite excellent lectures)
  1: https://videolectures.net/videos/mlss09uk_blei_tm?part=1
  2: https://videolectures.net/videos/mlss09uk_blei_tm?part=2

- Short video on LDA:
  https://www.youtube.com/watch?v=fCmIceNqVog

- Nice short article on the intuitions on LDA/Gibbs sampling:
  https://medium.com/analytics-vidhya/topic-modeling-using-lda-and-gibbs-sampling-explained-49d49b3d1045

- Coursera has a lot of small and large courses

THAT'S IT

Next week about vector semantics and embedding

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